

Program : Diploma in Information Technology	
Course Code : 5262	Course Title: Object Oriented Web Development
Semester : 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide basic concepts the Object-Oriented Concept of PHP.
- To develop web applications using Content Management Systems.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge about basic Web Development		Fundamentals of Web Development	3
Skills to design web pages with the help of PHP		Advanced Web Development	4

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate Object Oriented Concepts in PHP	14	Understanding
CO2	Illustrate Inheritance in PHP	14	Understanding
CO3	Illustrate Advanced Object-Oriented functionality in PHP, Error Handling in PHP, MVC Frameworks and CodeIgniter basics.	16	Applying
CO4	Illustrate CodeIgniter MVC Framework	14	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate Object Oriented Concepts in PHP		
MO1.01	Define class and Object.	1	Remembering
MO1.02	Define Polymorphism and Inheritance	1	Remembering
MO1.03	Define class in PHP	1	Remembering
MO1.04	Summarize the structure of a class in PHP	2	Understanding
MO1.05	Summarize the creation of objects in PHP	2	Understanding
MO1.06	Summarize access modifiers in PHP	2	Understanding
MO1.07	Summarize constructor and destructor in PHP	3	Understanding
MO1.08	Summarize Class constants in PHP.	2	Understanding
Contents: Define Class, Object, Polymorphism, and Inheritance. Define class in PHP, discuss structure of a class in PHP, create objects in PHP, constructor and destructor in PHP, Discuss access modifiers in PHP, Class constants in PHP.			
CO2	Illustrate Inheritance in PHP.		
M2.01	Summarize Inheritance in PHP	2	Understanding
M2.02	Summarize how to implement Inheritance in PHP	2	Understanding
M2.03	Summarize Control visibility through Inheritance with private and protected.	2	Understanding
M2.04	Define Overriding in PHP, summarize preventing Inheritance and overriding with final	3	Understanding

M2.05	Summarize Understand multiple Inheritance.	2	Understanding
M2.06	Summarize Implementing Interfaces in PHP.	3	Understanding
	Series Test – I	1	
Contents: Discuss Inheritance in PHP-Implement Inheritance in PHP- Control visibility through Inheritance with private and protected-Overriding-preventing Inheritance and overriding with final- Understand multiple Inheritance- Implementing Interfaces in PHP.			
CO3	Illustrate Advanced Object-Oriented functionality in PHP, Error Handling in PHP, MVC Frameworks and Code Igniter basics.		
M3.01	Summarize implementation of Static Methods	1	Understanding
M3.02	Summarize Checking class type and Type Hinting, Late Static Binding	1	Understanding
M3.03	Summarize Late Static Binding, Cloning Objects	1	Understanding
M3.04	Summarize Exception Handling concepts in PHP.	1	Understanding
M3.05	Summarize Discuss Exception class, User defined Exceptions.	1	Understanding
M3.06	Summarize MVC Architecture, Summarize MVC structure in PHP	3	Understanding
M3.07	Display a message using PHP MVC	1	Understanding
M3.08	Summarize CodeIgniter framework and its features	1	Understanding
M3.09	Installation of CodeIgniter	2	Applying
M3.10	Summarize basic concepts of CodeIgniter	2	Understanding
M3.11	Summarize database configuration in CodeIgniter	2	Applying
Contents: Advanced Object-Oriented functionality - Implement Static Methods, checking class type and Type Hinting, Late Static Binding, Cloning Objects. Discuss Exception Handling concepts in PHP, Discuss Exception class, User defined Exceptions. MVC Architecture, Discuss MVC structure in PHP, display a message using PHP MVC. CodeIgniter framework – CodeIgniter framework and its features, Installation of CodeIgniter, basic concepts of CodeIgniter, Configuration of CodeIgniter.			
CO4	Illustrate CodeIgniter MVC Framework		
M4.01	Summarize CodeIgniter Connecting to a Database, Insertion, updation, selection and deletion of records	3	Remembering
M4.02	Summarize Libraries	1	Understanding
M4.03	Summarize CodeIgniter Error Handling	2	Understanding
M4.04	Summarize CodeIgniter File Uploading	1	Understanding

M4.05	Summarize CodeIgniter Sending Email	1	Understanding
M4.06	Summarize CodeIgniter Form Validation	1	Applying
M4.07	Summarize CodeIgniter Session Management	1	Applying
M4.08	Summarize CodeIgniter Cookie Management	1	Applying
M4.09	Summarize CodeIgniter Flashdata and Tempdata	1	Applying
M4.10	Summarize CodeIgniter Page Caching and page redirection	2	Applying
	Series Test – II	1	
Contents: CodeIgniter database connection – CodeIgniter Libraries – CodeIgniter Error Handling –CodeIgniter File Uploading and sending email – CodeIgniter Form Validation – CodeIgniter session and Cookie Management –CodeIgniter Flashdata and Tempdata - CodeIgniter Page Caching and page redirection.			

Text / Reference

T/R	Book Title/Author
T1	PHP and MySQL web development -Fifth edition-Luke Welling, Laura Thomson
T2	Learn WordPress in Easy way-A beginner's Guide- Dr Rithesh Kumar
T3	CodeIgniter 3 Cookbook, by Rob Foster

Online Resources

SLNo	Website Link
1	https://www.javatpoint.com/php-mvc-architecture
2	https://www.studentstutorial.com/php/mvc/mvc-structure
3	https://www.tutorialspoint.com/codeigniter/index.htm